

Question Number	Answer	Additional guidance	Mark
2(a)(i)	A calculation in which: • conversion of actual length of scale into nm (1) • actual length divided by the given length (answer given to 2 significant figures and in standard form) (1)	<u>Example of calculation:</u> 35mm = 35 000 000 nm $35\,000\,000 \div 1200 = \times 2.9 \times 10^4$ accept +/- 1mm for length of bar	(2)

Question Number	Answer	Additional guidance	Mark
2(a)(ii)	An explanation that makes reference to two of the following points: • electron microscope (1) • due to high magnification (1)	ecf from 2(a)(i) applies Accept high resolution Accept converse for light microscope Accept 3D image if qualified as scanning electron microscope	(2)

Question Number	Answer	Additional guidance	Mark
2(b)(i)	An answer that makes reference to the following points: • presence of {(peptidoglycan) cell wall / circular chromosome(s) / DNA associated with histones / 70S ribosomes / RNA polymerase} (1) • absence of {nucleus / membrane bound organelles} (1)	Accept ether bonds	(2)

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that makes reference to three of the following points: • (analysis of) molecular evidence (1) • to identify similarities and differences in biological molecules / comparison of biological molecules (1) • (analysis of) phenotype (1) • to identify similarities and differences between the two microorganisms (phenotype) (1)	e.g. DNA, mRNA, proteins, enzymes e.g. cell structure / anatomical features, grow in different {habitats/ conditions} e.g. named similarities and differences in cell structure	(3)